Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/10/2018
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Project Title:	Eutectic Solidification in a Ternary Alloy		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
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Note this is a follow-on project for Dr. Ralph Napolitano's project,"Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy," which ended in 2014. Solidification processing of metallic alloys can produce a wide array of growth morphologies, which can be manipulated to access new structures, materials, and functionalities. The array of accessible structures is particularly wide in ternary and other higher-order systems, but the challenge of controlling the many variables that influence growth structures is immense and requires a better understanding of the interplay between the mechanisms of pattern formation and the influence of applied forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and

Task Description:	applied thermal field. The two principle objectives to be addressed are (i) to establish an analytical framework for 2D and 3D quantitative description of multiphase microstructures, and (ii) to employ the newly developed quantitative framework to investigate the dynamics of pattern formation in ternary eutectic alloys. The work will focus primarily on the invariant three-phase eutectic structure. Ground-based samples will be processed via a vertical Bridgman-style directional solidification furnace. Electron backscatter diffraction will be a major characterization tool to measure crystallographic orientation, texture, and misorientation relationships. Analysis over the length of the sample will provide insight into initial transient behavior, grain competition, and steady state growth. The relationship between geometric and crystallographic parameters, such as curvatures, orientations, and periodic ordering, will be used to quantify the morphological trajectory of the eutectic and identify fundamental principles of pattern formation. In order to validate the experimental correlations and analytical models developed from the findings of the ground-based work, carefully selected flight-based directional solidification experiments will be conducted using the European Space Agency (ESA) Materials Science Lab (MSL) and the Low Gradient Furnace (LGF). The project operates in collaboration with the European Space Agency's project titled "Solidification along a Eutectic Path in Ternary Alloys" (SETA), which has already accomplished important work using the Al-Ag-Cu alloy system. Significant findings of the SETA project to date have been in the areas of pattern development and the role of fault structure for both univariant and invariant eutectic compositions, the role of crystal orientation and "locked" grains, and development of nonplanar growth modes. Facilitating open scientific interaction between the Principal Investigator (PI) and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. The planning and performance of flight-based solidification experiments is being done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled, multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is far-reaching, given the broad range of multiphase structures that can be fabricated through solidification processing, and is not limited to metals and alloys. Indeed, the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
	Specific areas of focus during Year 4 have been (i) EBSD (electron backscatter diffraction) analysis to determine the most common orientation relationships in the Al-Ag-Cu three phase eutectic, (ii) tomography for 3D characterization of the three phase eutectic microstructures, (iii) the evolution of Al-Ag-Cu eutectics due to solid state diffusion, and (iv) final development and testing of code to perform automated image analysis of the three-phase eutectic structures. 1. CRYSTALLOGRAPHIC ORIENTATION RELATIONSHIPS IN TERNARY EUTECTIC AL-AG-CU The microstructure of ternary eutectic Al-Al2Cu-Ag2Al arranges in several patterns of three solid phases during directional solidification. One key question for understanding the behavior of this system is if and how the patterns depend on crystal orientation relationships (ORs) between the solid phases. In order to study the correlation between the ORs and the evolving patterns for different process conditions, electron backscatter diffraction (EBSD) was performed on samples of directionally solidified ternary eutectic Al-Ag-Cu which had been processed with different solidification velocities and temperature gradients. Data for this study was a compilation of results from the Principal Investigator (PI) and other SETA team members. The results show that characteristic ORs occur, influencing the type of the evolving pattern, the alignment of the phases, and the degree of order. Chain-like structures are the most commonly observed pattern reported in this ternary eutectic. Pole figures show that the (110) plane of the Al2Cu is involved as the common plane between both Al2Cu and Al and the perpendicular plane between Al2Cu and Ag2Al, so it seems as if this plane and its 4-fold symmetry is driving the pattern formation. The lattice parameters of the planes were calculated and found to match fairly closely, indicating a semi-coherent interface and therefore likely a low interfacial energy. This matches well with previous findings from phase field modeling that show this pattern becomes more aligned with increasing solidification distance. A number of grains (>20) with a variety of microstructural patterns were analyzed. Eighty percent of the chain-like patterns had the OR described above. In other regions, another orientation relationship that was observed included the two orthogonal Al2Cu plane of the {130} family, orientated parallel to the {0001} basal plane of the Ag2Al. In this situation, the Al phase was found to have many different orientations, with apparently almost any low index plane involved. In this OR, the structure appears more disordered and the phase boundaries much less aligned. 2. TOMOGRAPHY FOR 3D ANALYSIS OF TERNARY EUTECTIC GROWTH The first component of the tomography analysis performed in the past year was the analysis of several samples created and imaged by SETA partner DLR (German space agency), and analyzed in conjunction with the phase field modeling group at Karlsruhe Institute of Technology. The tomography was performed at the European Synchrotron Radiation Facility in Grenoble, France. The tomography data was converted into a full 3D reconstruction for two pieces of metal taken from two different grains of the same initial sample, one with a more aligned pattern and one with a less aligned pattern. A novel form of graph-based analysis was used to identify regions in the samples where individual rods split, merged, terminated, or nucleated and allowing the frequency of these events to be graphed over the length of the samples. Although nucleation in directionally solidified eutectics is controversial, the tomography data seems to show quite clearly in a few instances that this is occurring. More investigation, of course, is needed. The main conclusions of this work are (i) For lower growth velocities, the number of microstructural adjustment events is lower. This is also valid if the number of events are scaled by the number of rods in the corresponding cross section. (ii) Despite several hours of steady growth conditions, the microstructure continues to adjust by the different events, leading to a three-dimensional highly connected structure of the rods and lamellae. (iii) The microstructure adjustment for constant growth velocities is accomplished primarily by split and merge events for all solid phases. (iv) For a

Task Progress:	decrease in the velocity, the coarsening of the microstructure is a result of rod termination and overgrowing. (v) For splitting and merging, the Al2Cu phase shows the highest number of events. However, for overgrowing and assumed nucleation, the Ag2Al phase shows the highest number of events. (vi) During the whole growth period, the tomography data occasionally shows the appearance of new rods which seem to be nucleation. Compared to the other events, it is relatively rare. (vii) Both regions of the sample that were studied show similar behavior and trends regarding microstructural adjustment events during steady growth and in response to a velocity change. However, noticeable differences in the number of events are observed in regions with different microstructural patterns. This work also demonstrates that the application of graph-based analysis to three-dimensional microstructure data obtained with synchrotron tomography provides new insights into the complex physical interactions of the microstructure adjustment process in three dimensions. Additional tomographic data was collected at the Advanced Photon Source at Argonne National Lab from samples processed at UAB (University of Alabama at Birmingham). Two regions (diameter = 0.5 mm, length = 6-7 mm) were machined from a sample obtained from various processing conditions and then subjected to tomography. Analysis of the enormous quantity of data collected during the tomography session is just beginning. 3. SOLID STATE CHANGES IN AL-AG-CU TERNARY EUTECTIC One challenge in studying the ternary eutectic system Al-Ag-Cu is a relative large solubility shift of the chemical element Ag in the Al phase after the solidification, which influences both the phase fractions and composition. The effects of these changes on the eutectic pattern themselves are not yet understood. To study these effects, directionally solidified samples of ternary eutectic Al-Ag2Al-Al2Cu were quenched. Then, the microstructure evolution in cross section at the quenched solid-liquid interface due to systematic annealing just below the eutectic temperature was recorded and quantitatively analyzed. This mimics the process that occurs naturally during cooling in the directional solidification furnace. The advantage of this approach is that the exact same region of microstructure can be studied. The results show that the phase fraction changes in the micrographs are in good accordance with the theoretical prediction from the thermodynamic Calphad database. Different patterns show distinct responses to the annealing. The chain-like structure shows less of a pattern change over the course of 32 total hours of annealing, compared to the less aligned grain. 4. CREATION AND OPTIMIZATION OF AUTOMATED IMAGE ANALYSIS TOOLS Quantification of the three-phase eutectic microstructure in AlAgCu is difficult because of the variability of the structure. Methods for extracting various quantitative parameters from the micrographs have been underway since the beginning of the project and made significant progress during this year. All work was done in Matlab. With this program, micrographs perpendicular to the growth direction are segmented and analyzed. For this, the micrographs and the necessary parameters are first read into the program. To be able to analyze large datasets (for example from tomography experiments) in an automatic manner, it is either possible to read single files or whole folders containing multiple micrographs. To improve the segmentation it is possible to apply sharpening algorithms with user-defined parameters, controlling the radius and amount of the sharpening stencil. After reading the data and sharpening, the images are segmented with a multi-threshold image segmentation algorithm based on Otsu's method. Even for images which are only slightly out of focus and hence appear blurred, boundaries between the black and the white phase are interpreted as gray phase. A direct quantitative analysis of these images would therefore result in wrong quantitative results. Also pixel errors lead to a false identification in the segmented images. To overcome this challenge, image correction algorithms are applied in the next step. Different methods based on the neighboring pixels, systematic dilation and erosion of the black and white particles, and the minimum particle size are implemented. The order and extent to which these methods are applied is flexible, as specified by the user. After segmentation, the micrographs are quantitatively accessed by determining the phase fractions of the three solid phases, the length of the interfaces between the particles, the angular orientation of these interphases, the lamellar spacings, the number of nearest neighbors, and the average particle size. For the calculation of the interface length an interpolation with cubic splines is applied which allows to determine the length rotationally invariant. The interpolated surface is also used to calculate the normal vectors for the angular orientation. The lamellar spacings are calculated by determining the barycenter distances between particles from the same phase. By taking into account the phase between the particles, different lamellar spacings are encountered, which is necessary to characterize the patterns. The calculated values are stored in a CSV (comma-separated value) file, which allows an efficient statistical analysis of the results. To simplify documentation and reproducibility of the results, the input parameters are also stored in file together with the results. The code was originally developed for analysis of Al-Ag-Cu, but its applicability has also been demonstrated for Al-Cu-Mg. Analysis of data from the tomography work described above shows that the program can also perform efficient analysis of very large data sets, and that good results are possible despite blurring of the initial images.
Bibliography Type:	Description: (Last Updated: 06/05/2023)
Articles in Peer-reviewed Journals	Steinmetz P, Hötzer J, Kellner M, Genau A, Nestler B. "Study of pattern selection in 3D phase field simulations during the directional solidification of ternary eutectic Al-Ag-Cu." Computational Materials Science. 2018 Jun 1;148:131-40. https://doi.org/10.1016/j.commatsci.2018.02.040 , Jun-2018